

11. Schmitt Trigger using OP-Amp

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1 Aim of the Experiment

To design a Schmitt trigger using IC 741 for a specified Lower Threshold Point (LTP) and Upper Threshold Point (UTP) and to plot the hysteresis curve.

2 Theory

A Schmitt trigger is a comparator circuit with hysteresis. It is often called a squaring circuit because it converts an irregular shaped input waveform into a square wave. The output voltage changes its state every time the input voltage crosses certain threshold levels.

The input voltage at which the output switches from positive saturation ($+V_{sat}$) to negative saturation ($-V_{sat}$) is called the Upper Triggering Point or Upper Threshold Point (UTP). Similarly, the input voltage at which the output switches back from $-V_{sat}$ to $+V_{sat}$ is called the Lower Triggering Point or Lower Threshold Point (LTP).

These threshold voltages are determined by the voltage divider network formed by resistors R_1 and R_2 . When the output is at $+V_{sat}$, the voltage across R_1 gives the UTP:

$$V_{UTP} = \frac{R_1}{R_1 + R_2}(+V_{sat})$$

When the input voltage exceeds V_{UTP} , the output switches to $-V_{sat}$. At this state, the voltage across R_1 gives the LTP:

$$V_{LTP} = \frac{R_1}{R_1 + R_2}(-V_{sat})$$

When the input voltage drops below V_{LTP} , the output switches back to $+V_{sat}$. The difference between the upper and lower threshold points creates a hysteresis loop.

3 Procedure

1. Verify whether the op-amp IC is in good working condition by using it as a voltage follower.
2. Set up the circuit on the breadboard as per the circuit diagram.

3. Apply a sinusoidal input signal (1 kHz, 15 V peak) from the function generator.
4. Connect the input to one channel and the output to the other channel of the CRO. Observe both waveforms simultaneously.
5. Design the circuit for different values of LTP and UTP by changing the values of resistors R_1 and R_2 .
6. Measure the upper and lower threshold voltages from the CRO screen for each combination of resistors.
7. Observe the transfer characteristics by switching the CRO to XY mode and plot the hysteresis curve.

4 Circuit Diagram

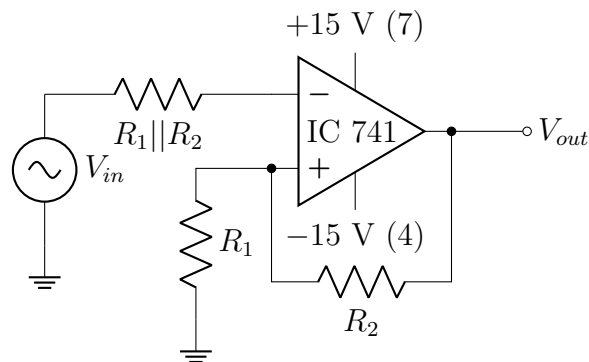


Figure 1: Circuit diagram of Schmitt Trigger using IC 741

5 Waveforms and Transfer Characteristics

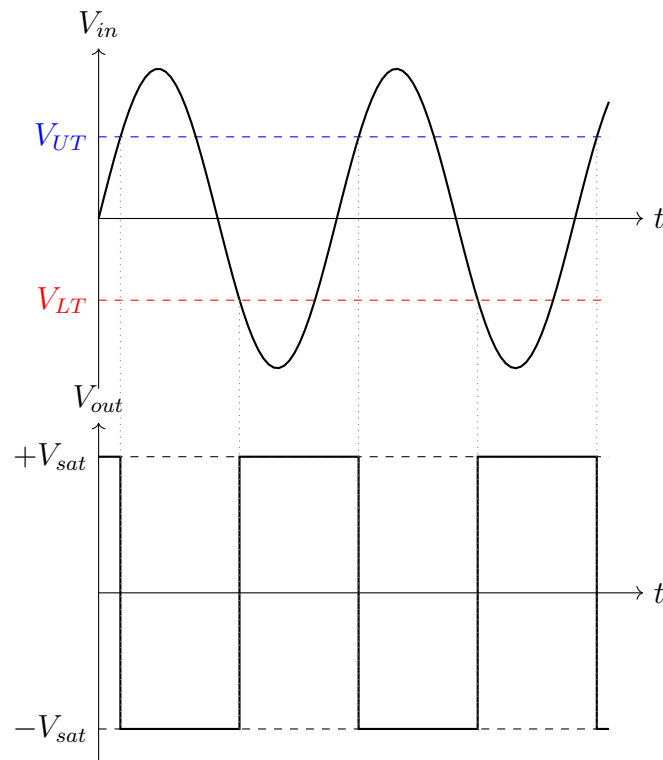


Figure 2: Input and Output waveforms of the Schmitt Trigger

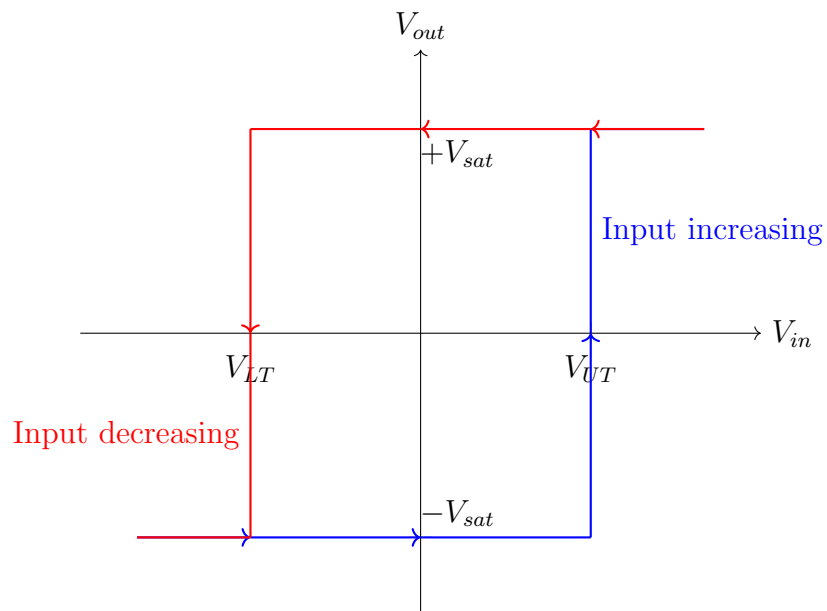


Figure 3: Transfer characteristics (Hysteresis curve) of the Schmitt Trigger

6 Tabular Column

R_1	R_2	Designed (V)		Measured (V)		H_{Theo} (V)	H_{meas} (V)
		UTP	LTP	UTP	LTP		
5.4 k Ω	10 k Ω	5.5	-5.5	5.6	-5.6	5.5	5.75
1.5 k Ω	10 k Ω	2.0	-2.0	2.9	-2.3	2.0	2.15
10 k Ω	47 k Ω	2.7	-2.7	3.5	-3.1	2.7	3.8
10 k Ω	56 k Ω	2.3	-2.3	3.2	-2.7	2.3	3.4
1.5 k Ω	5.4 k Ω	3.33	-3.33	4.0	-3.4	3.33	4.0

Table 1: Observations for Schmitt Trigger Thresholds

7 Calculations

The theoretical threshold voltages are calculated using the formula:

$$V_{UT} = \frac{R_1}{R_1 + R_2}(+V_{sat})$$

$$V_{LT} = \frac{R_1}{R_1 + R_2}(-V_{sat})$$

Assuming the practical saturation voltage of the op-amp is approximately ± 15.7 V (which corresponds to the expected values):

For $R_1 = 5.4$ k Ω and $R_2 = 10$ k Ω :

$$V_{UT} = \frac{5.4}{5.4 + 10} \times 15.7 \approx 5.5 \text{ V}$$

$$V_{LT} = \frac{5.4}{5.4 + 10} \times (-15.7) \approx -5.5 \text{ V}$$

8 Results

The Schmitt trigger circuit was successfully designed and set up using the IC 741 operational amplifier. The Upper Threshold Point (UTP) and Lower Threshold Point (LTP) were measured for various combinations of resistors R_1 and R_2 . The measured threshold values were found to be close to the theoretical design values. The hysteresis curve was also successfully observed on the CRO in XY mode.

9 Error Calculation

Taking the first set of readings ($R_1 = 5.4$ k Ω , $R_2 = 10$ k Ω): Theoretical $V_{UT} = 5.5$ V
Measured $V_{UT} = 5.6$ V

$$\text{Percentage Error} = \frac{|5.6 - 5.5|}{5.5} \times 100\% \approx 1.81\%$$

10 Discussion

In this experiment, we studied the operation of a Schmitt trigger using an operational amplifier. We learned that the Schmitt trigger acts as a comparator with built-in hysteresis. This hysteresis property is extremely useful in practical circuits as it prevents noise in the input signal from causing multiple false triggers at the output.

The theoretical values of the upper and lower threshold voltages were calculated and compared with the measured values. The measured threshold voltages were slightly higher than the theoretical ones in some cases. This difference arises because the exact value of the saturation voltage V_{sat} of the op-amp is not perfectly ideal and depends on the specific IC and the applied power supply. Furthermore, the tolerance limits of the resistors used also contribute to the minor variations between designed and measured values.

When plotting the transfer characteristics in XY mode on the CRO, we could clearly observe the rectangular hysteresis loop, which demonstrated the memory-like property of the circuit. The width of this loop corresponds to the hysteresis voltage.

11 References

1. Lab Manual for Advanced Physics Lab II, IISER TVM.
2. Boylestad, R. L., & Nashelsky, L., *Electronic Devices and Circuit Theory*, Pearson.

12 Lab Notes (Observations)

The following handwritten notes and observations were recorded during the laboratory session.

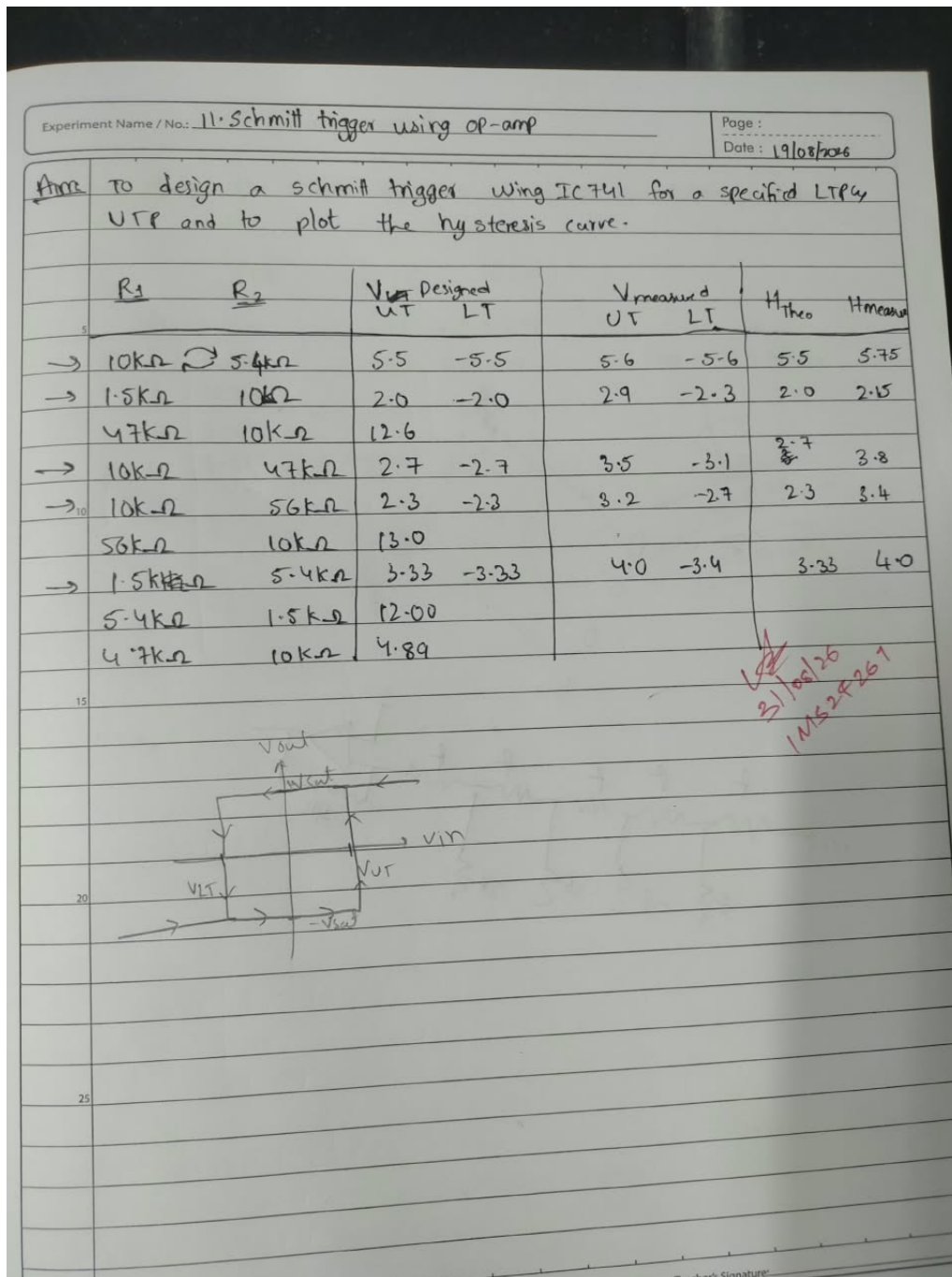


Figure 4: Lab Notes for Schmitt Trigger Observations

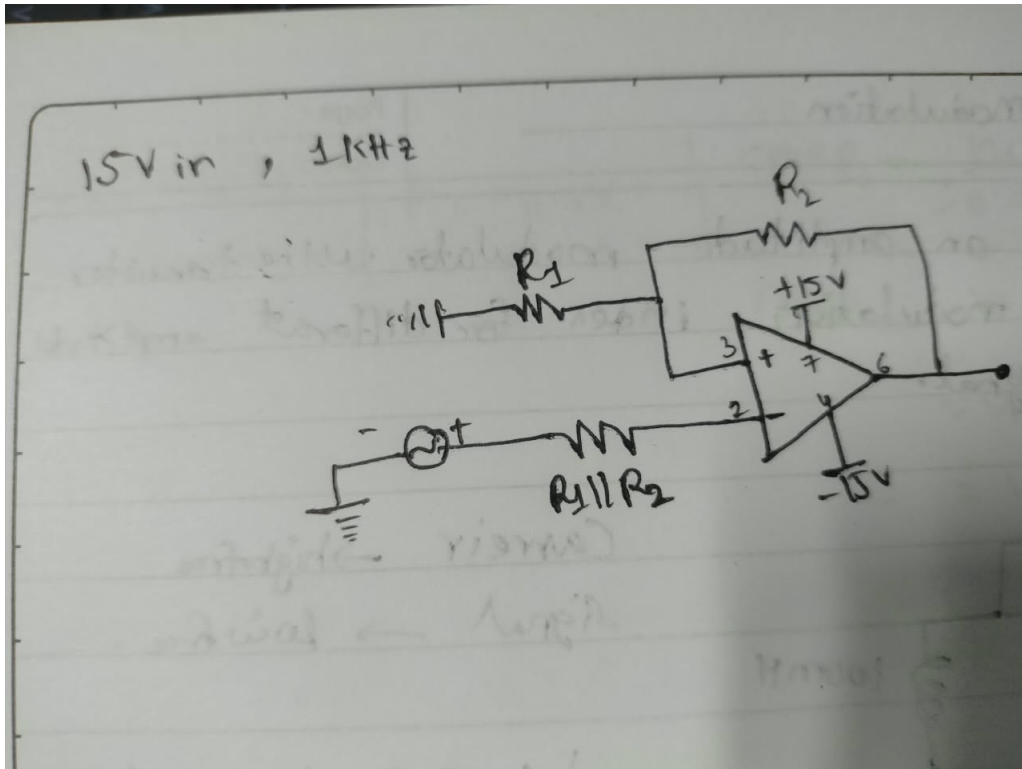


Figure 5: Handwritten Circuit Diagram for Schmitt Trigger

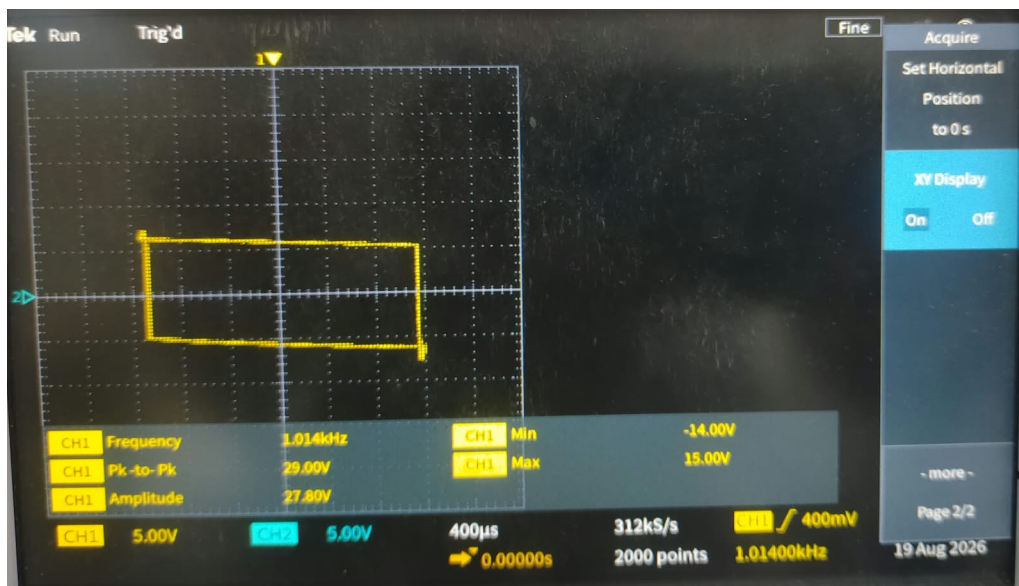


Figure 6: Oscilloscope trace showing the observed hysteresis curve in XY mode

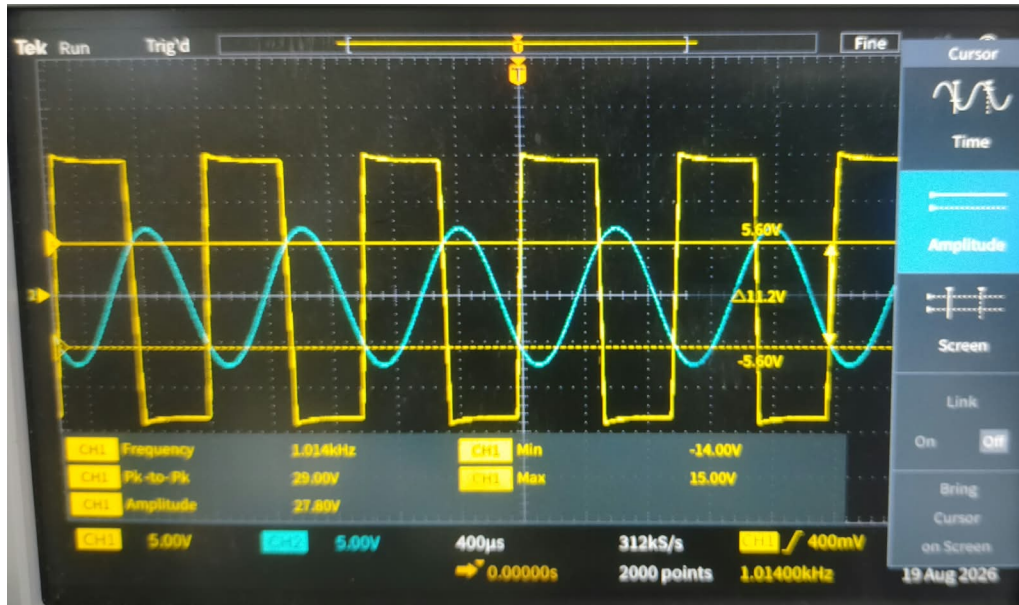


Figure 7: Oscilloscope trace showing the input sine wave and the output square wave